M. HANIF AL FAIZ

Informatics Undergraduate • Software & Intelligent Systems Prototyping

Telkom University Purwokerto • GitHub: github.com/hanif-12-01

PROFESSIONAL SUMMARY

Undergraduate Informatics student at Telkom University Purwokerto focusing on software engineering, intelligent systems, and civic technology. Experience includes building verifiable decision-support prototypes, client-side machine learning inference pipelines, and full-stack web applications. Recognized as a national competition finalist and podium recipient with active contributions to faculty research, student creativity initiatives, and university student governance.

EDUCATION

Telkom University Purwokerto — **Purwokerto, Indonesia**

Bachelor of Informatics (S1 Candidate) — *Expected: 2027*

- **Academic Status:** Current Cumulative GPA: **3.66 / 4.00** | Current Status: **Semester 5 (In Progress)**.
- **Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, Software Engineering, Database Systems, Computer Networks, Discrete Mathematics, Object-Oriented Programming.

TECHNICAL CAPABILITIES

Programming Languages:	C, C++, Python, Java, Golang
Software Development:	Software Engineering Fundamentals, Application Development, Requirement Analysis, Software Testing & Validation
Developer Tools & Platforms:	Visual Studio Code, Git, GitHub, MySQL, Docker, Figma, XAMPP, Android Studio, Visual Studio
Foundations & Logic:	Algorithmic Thinking, Programming Logic, Problem Solving, Basic Data Structures, Debugging & Code Tracing
Communication & Leadership:	Technical Communication, Public Speaking, Conference / Event Presentation, Team Leadership, Team Coordination, Project Coordination, Collaborative Problem Solving

KEY ENGINEERING PROJECTS

WattWise AI — Electricity-Cost Decision-Support Platform — 2026

Next.js 16, React 19, TypeScript, PostgreSQL, Drizzle ORM, ONNX Runtime Web, Tesseract.js | Working MVP

- Developed a working MVP decision-support platform tailored for Indonesian small property operators (kos, boarding houses) and energy-intensive SMEs to optimize utility expenses.
- Implemented client-side N-BEATS load forecasting executing directly in the browser via ONNX Runtime Web, eliminating remote GPU server dependency.
- Built privacy-preserving in-browser optical character recognition using Tesseract.js for physical meter photo scanning and tariff tier calculation.
- Recognized as a **Top 10 Finalist** in the PLN Innovation & Clean Energy (ICE) 2026 Startup Competition.

CIVICTWIN Semarang — Civic Intelligence Proof of Concept — 2026

React 18, TypeScript, Vite, Tailwind CSS, Leaflet, React Router | Competition Proof of Concept (Demo Data)

- Developed a civic intelligence proof of concept for Semarang linking district-priority mapping, citizen feedback, simulated OPD follow-up, and policy-scenario exploration.
- Implemented a browser-based demo using localStorage, mock reports, prototype district markers, and rule-based CivicSense guidance.
- Awarded **3rd Place** at the UNITY UNY 2026 Smart City Competition.

Purwokerto Intelligence Layer — Smart City Decision-Support Prototype — 2026

React 19, TypeScript, Vite, Tailwind CSS, Leaflet, React Router | Experimental Material (Simulated Data)

- Developed an interactive Smart City decision-support prototype covering Purwokerto's four subdistricts to model civic indicators and spatial urban analytics using simulated urban indicator datasets.
- Implemented a Multi-Criteria Decision Making (MCDM) weighted formula enabling dynamic policy weighting across public service, economic, and infrastructure indicators.
- Engineered geospatial GIS visualization with Leaflet and GeoJSON subdistrict boundaries, citizen feedback logging, and policy brief drafting modules.

Fuzzy Rain Risk Index — AI Coursework Prototype — 2026

Python, Jupyter Notebook, NumPy, Pandas, Matplotlib, Fuzzy Logic | AI Coursework Project

- Implemented both Fuzzy Mamdani (centroid defuzzification) and Fuzzy Sugeno (weighted average) inference models entirely from scratch in Python without external fuzzy libraries.
- Formulated custom linguistic variables and membership functions applied to Indonesian hourly meteorological records to estimate 6-hour outdoor rain risk.

HONORS & COMPETITION OUTCOMES

3rd Place — Smart City Competition, UNITY UNY (CIVICTWIN Semarang) — **2026**

Top 10 Finalist — PLN Innovation & Clean Energy (ICE) Startup Competition, PT PLN (Persero) — **2026**

Top 5 Finalist — GenBI Business Plan Competition, Generasi Baru Indonesia — **2026**

Grant Recipient — YIA KPP Mining Project Grant, Yayasan Inspirasi Anak Bangsa & PT Kalimantan Prima Persada **2024**

Competition Participant — GEMASTIK National ICT Competition (Smart City & ICT Business Divisions) **2024**

ACADEMIC RESEARCH & PRESENTATIONS

Faculty of Informatics Research Grant **2025**

Student Research Contributor *Telkom University Purwokerto*

- Collaborated with faculty researchers on data collection, feature extraction, and algorithm execution benchmarking for grant evaluation.

Program Kreativitas Mahasiswa (PKM) Initiatives **2025**

Proposal Contributor & Developer *Faculty-Sponsored*

- Contributed to two faculty-supported PKM initiatives through proposal development and technical prototyping.

PKM Data Science Academic Seminar **2025**

Session Speaker / Presenter *Academic Workshop*

- Presented statistical data models and empirical research findings to peer student cohorts during an academic departmental seminar.

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

Responsive Program — Informatics Student Association

Project Lead / Organizing Committee Chair *Telkom University Purwokerto*

- Led the student organizing committee coordinating orientation and technical onboarding activities for incoming Informatics students.

Badan Eksekutif Mahasiswa (BEM KEMA)

Secretary *Telkom University Purwokerto*

- Managed organizational documentation, administrative compliance, and correspondence for the university student executive board.